

Nutritional Values of Foods

Condensed food-composition reference compiled from *Human Nutrition: 2020 Edition* (University of Hawai'i at Manoa). Values are per stated serving; %DV = percent Daily Value.

ENERGY & MACRONUTRIENT REFERENCE

Energy yield per gram

Macronutrient	kcal / g
Carbohydrate	4
Protein	4
Fat (lipid)	9
Alcohol	7

Water content of foods (% water)

% Water	Example foods
90–99	Nonfat milk, cantaloupe, strawberries, watermelon, lettuce, cabbage, celery, spinach, squash
80–89	Fruit juice, yogurt, apples, grapes, oranges, carrots, broccoli, pears, pineapple
70–79	Bananas, avocados, cottage cheese, ricotta, baked potato, shrimp
60–69	Pasta, legumes, salmon, chicken breast
50–59	Ground beef, hot dogs, steak, feta cheese
40–49	Pizza
30–39	Cheddar cheese, bagels, bread
20–29	Pepperoni, cake, biscuits
10–19	Butter, margarine, raisins
1–9	Walnuts, dry-roasted peanuts, crackers, cereals, pretzels, peanut butter
0	Oils, sugars

CARBOHYDRATES, SODIUM & BEVERAGES

Carbohydrates in foods (g per serving)

Food (serving)	Total	Sugar	Fiber	Added
Banana (1 med)	27	14.4	3.1	0
Lentils (1 c.)	40	3.5	16.0	0
Snap beans (1 c.)	8.7	1.6	4.0	0
Green pepper (1 med)	5.5	2.9	2.0	0
Corn tortilla (1)	10.7	0.2	1.5	0
Bread, wheat bran (1 sl)	17.2	3.5	1.4	3.4
Bread, rye (1 sl)	15.5	1.2	1.9	1.0
Bagel, plain (1 med)	53	5.3	2.3	4.8
Brownie (1 sq)	36	20.5	1.2	20.0
Oatmeal cookie (1 oz)	22.3	12.0	2.0	7.7
Cornflakes (1 c.)	23	1.5	0.3	1.5
Pretzels (10 twists)	47	1.3	1.7	0
Popcorn, homemade (100 g)	58	0.5	10.0	0
Skim milk (1 c.)	12	12.0	0	0
Half-and-half (1 Tbs)	0.65	0.02	0	0
Cream substitute (1 tsp)	1.0	1.0	0	1.0
Cheddar cheese (1 sl)	1.3	0.5	0	0
Yogurt w/ fruit (6 oz)	32.3	32.3	0	19.4
Caesar dressing (1 Tbs)	2.8	2.8	0	2.4

Sodium content of selected foods

Food	Serving	Na (mg)
Breads, all types	1 oz.	95–210
Rice Chex cereal	1¼ c.	292
Raisin Bran cereal	1 c.	362
Frozen pizza, cheese	4 oz.	450–1200
Frozen vegetables	½ c.	2–160
Salad dressing, reg.	2 Tbsp.	110–505
Salsa	2 Tbsp.	150–240
Soup, tomato	8 oz.	700–1260
Potato chips	1 oz.	120–180
Tortilla chips	1 oz.	105–160
Pork	3 oz.	59
Chicken	½ breast	69
Chicken fast-food dinner	—	2243
Chicken noodle soup	1 c.	1107
Dill pickle	1	928
Soy sauce	1 Tbsp.	1029
Canned corn	1 c.	384
Baked beans, canned	1 c.	856
Hot dog	1	639
Burger, fast-food	1	990
Steak	3 oz.	55
Canned tuna	3 oz.	384
Fresh tuna	3 oz.	50
Dry-roasted peanuts	1 c.	986
American cheese	1 oz.	406
Tap water	8 oz.	12

Calories in beverages

Beverage	oz	kcal
Soda	12	124–189
Bottled sweet tea	12	129–143
Orange juice	12	157–168
Tomato/veg juice	12	80
Whole milk	12	220
Nonfat milk	12	125
Soy milk	12	147–191
Coffee, black	12	0–4
Coffee, w/ cream	12	39–43
Caffe latte, whole	12	200
Sports drink	12	94
Beer	12	153
White wine	5	122

Caffeine content

Beverage / food	mg
Starbucks Grande coffee (16 oz)	380
Plain brewed coffee (8 oz)	102–200
Espresso (1 oz)	30–90
Decaf coffee (8 oz)	3–12
Tea, brewed (8 oz)	40–120
Green tea (8 oz)	25–40
Coca-Cola Classic (12 oz)	35
Dr. Pepper (12 oz)	44
Jolt Cola (12 oz)	72
Mountain Dew (12 oz)	54
Mtn Dew MDX (12 oz)	71
Pepsi-Cola (12 oz)	38
Red Bull (8.5 oz)	80
Full Throttle (16 oz)	144
Monster Energy (16 oz)	160
Spike Shooter (8.4 oz)	300

FAT-SOLUBLE VITAMINS (A, D, E, K)

Vitamin A (IU)

Food	Serv	IU	%DV
Beef liver	3 oz	27,185	545
Chicken liver	3 oz	12,325	245
Milk, skim	1 c.	500	10
Milk, whole	1 c.	249	5
Cheddar cheese	1 oz	284	6

Alpha- & Beta-Carotene (mg)

Food	Serv	β-car	α-car
Pumpkin, canned	1 c.	17.0	11.7
Carrot juice	1 c.	22.0	10.2
Carrots, cooked	1 c.	13.0	5.9
Carrots, raw	1 med	5.1	2.1
Winter squash, baked	1 c.	5.7	1.4
Collards, cooked	1 c.	11.6	0.2
Tomato	1 med	0.55	0.10
Tangerine	1 med	0.13	0.09
Peas, cooked	1 c.	1.20	0.09

Vitamin D (IU)

Food	Serv	IU	%DV
Swordfish	3 oz	566	142
Salmon	3 oz	447	112
Tuna, canned in water	3 oz	154	39
OJ, vit-D fortified	1 c.	137	34
Milk, vit-D fortified	1 c.	115–124	29–31
Margarine, fortified	1 Tbsp	60	15
Sardines, canned in oil	2 ea	46	12
Beef liver	3 oz	42	11
Egg, large	1 ea	41	10

Vitamin E (mg)

Food	Serv	mg	%DV
Sunflower seeds	1 oz	7.4	37
Almonds	1 oz	6.8	34
Sunflower oil	1 Tbsp	5.6	28
Hazelnuts	1 oz	4.3	22
Peanut butter	2 Tbsp	2.9	15
Peanuts	1 oz	2.2	11
Corn oil	1 Tbsp	1.9	10
Kiwi	1 med	1.1	6
Tomato	1 med	0.7	4
Spinach, raw	1 c.	0.6	3

Vitamin K (mcg)

Food	Serv	mcg	%DV
Broccoli	½ c.	160	133
Cabbage	½ c.	56	47
Asparagus	4 spears	34	28
Spinach	½ c.	27	23
Green peas	½ c.	16	13
Ham	3 oz	13	11
Cheese	1 oz	10	8
Ground beef	3 oz	6	5
Orange	1 ea	1.3	1
Bread	1 slice	1.1	<1

WATER-SOLUBLE VITAMINS (C, B-complex)

Vitamin C (mg)

Food	Serv	mg	%DV
Sweet red pepper	½ c. raw	95	158
Orange juice	6 oz	93	155
Strawberries	1 c.	85	164
Grapefruit juice	6 oz	70	117
Orange	1 med	70	117
Cauliflower	1 c. boiled	55	86
Broccoli	½ c. ckd	51	65
Romaine lettuce	2 c.	28	47
Tomato	1 med	17	28
Potato, baked	1 med	17	28

Thiamin B1 (mg)

Food	Serv	mg	%DV
Cereals, fortified	1 serv	1.5	100
White rice, enriched	½ c.	1.4	73
Pork chop, broiled	3 oz	0.4	27
Black beans, boiled	½ c.	0.4	27
Tuna, cooked	3 oz	0.2	13
Brown rice (unenr.)	½ c.	0.1	7
Whole wheat bread	1 slice	0.1	7
2% milk	8 oz	0.1	7
Cheddar cheese	1½ oz	0	0
Apple, sliced	1 c.	0	0

Riboflavin B2 (mg)

Food	Serv	mg	%DV
Beef liver	3 oz	2.9	171
Cereals, fortified	1 serv	1.7	100
Instant oats, fortified	1 c.	1.1	65
Plain yogurt, fat-free	1 c.	0.6	35
2% milk	8 oz	0.5	29
Beef tenderloin steak	3 oz	0.4	24
Portabella mushrooms	½ c.	0.3	18
Almonds, dry roasted	1 oz	0.3	18
Egg, scrambled	1 lg	0.2	12
Quinoa	1 c.	0.2	12
Salmon, canned	3 oz	0.2	12
Spinach, raw	1 c.	0.1	6
Brown rice	½ c.	0	0

Niacin B3 (mg)

Food	Serv	mg	%DV
Cereal, fortified	1 c.	20	100
Turkey	3 oz	10.0	50
Tuna	3 oz	8.6	43
Salmon	3 oz	8.5	42.5
Chicken	3 oz	7.3	36.5
Cereal, unfortified	1 c.	5	25
Beef, 90% lean	3 oz	4.4	22
Peanuts	1 oz	3.8	19
Whole wheat bread	1 slice	1.3	6.5
Coffee	8 oz	0.5	2.5

Pantothenic acid B5 (mg)

Food	Serv	mg	%DV
Sunflower seeds	1 oz	2	20
Fish, trout	3 oz	1.9	19
Yogurt, plain nonfat	8 oz	1.6	16
Lobster	3 oz	1.4	14
Avocado	½ fruit	1	10
Sweet potato	1 med	1	10
Milk	8 oz	0.87	8.7
Egg	1 lg	0.7	7
Orange	1 whole	0.3	3
Whole wheat bread	1 slice	0.21	2.1

Biotin B7 (mcg)

Food	Serv	mcg	%DV
Eggs	1 lg	10	33.3
Salmon, canned	3 oz	5	16.6
Pork chop	3 oz	3.8	12.6
Sunflower seeds	¼ c.	2.6	8.6
Sweet potato	½ c.	2.4	8
Almonds	¼ c.	1.5	5
Tuna, canned	3 oz	0.6	2
Broccoli	½ c.	0.4	1.3
Banana	½ c.	0.2	0.6

Vitamin B6 (mg)

Food	Serv	mg	%DV
Chickpeas	1 c.	1.1	55
Tuna, fresh	3 oz	0.9	45
Salmon	3 oz	0.6	30
Potatoes	1 c.	0.4	20
Banana	1 med	0.4	20
Ground beef patty	3 oz	0.3	10
White rice, enriched	1 c.	0.1	5
Spinach	½ c.	0.1	5

Folate (mcg DFE)

Food	Serv	mcg	%DV
Cereals, fortified	¾ c.	400	100
Beef liver	3 oz	215	54
Spinach	½ c.	131	33
White rice, enriched	½ c.	90	23
Asparagus	4 spears	85	20
Avocado	½ c.	59	15
Broccoli	2 spears	45	10
White bread, enriched	1 slice	43	11
Orange juice	6 oz	35	9
Egg	1 lg	22	6

Vitamin B12 (mcg)

Food	Serv	mcg	%DV
Clams	3 oz	84.1	1,402
Salmon	3 oz	4.8	80
Tuna, canned	3 oz	2.5	42
Cereals, fortified	1 serv	1.5	25
Beef, top sirloin	3 oz	1.4	23
Milk, lowfat	8 oz	1.2	18
Yogurt, lowfat	8 oz	1.1	18
Cheese, Swiss	1 oz	0.9	15
Egg	1 lg	0.6	10

Choline (mg)

Food	Serv	mg	%DV
Egg	1 lg	147	27
Soybeans	½ c.	107	19
Chicken breast	3 oz	72	13
Mushrooms, shiitake	½ c.	58	11
Potatoes	1 lg	57	10
Kidney beans	½ c.	45	8
Peanuts	¼ c.	24	4
Brown rice	1 c.	19	3

MAJOR MINERALS

Calcium (mg) & absorption

Food	Serv	mg	Abs%
Mozzarella, part skim	1.5 oz	333	32
Sardines, w/ bones	3 oz	325	27
Cheddar cheese	1.5 oz	303	32
Yogurt, low fat	8.5 oz	300	32
Milk, nonfat	8.5 oz	300	32
Soy milk, Ca-fortified	8.5 oz	300	24
OJ, Ca-fortified	8.5 oz	300	36
Tofu, w/ Ca sulfate	½ c.	258	31

Phosphorus (mg)

Food	Serv	mg	%DV
Salmon	3 oz	315	32
Yogurt, nonfat	8 oz	306	31
Turkey, light meat	3 oz	217	22
Beef	3 oz	179	18
Lentils	½ c.	178	18
Almonds	1 oz	136	14
Chicken, light meat	3 oz	135	14
Mozzarella	1 oz	131	13
Peanuts	1 oz	108	11
Egg	1 lg	86	9
Whole wheat bread	1 slice	68	7
Cola drink	12 oz	41	4
Bread, enriched	1 slice	25	3

Sulfur amino acids (mg)

Food	Serv	Met	Cys
Turkey breast	1 breast	7102	2442
Brazil nuts	1 c.	1495	407
Canned tuna	1 c.	1259	456
Swiss cheese, diced	1 c.	1035	383
Canned green peas	1 can	257	0.172
Egg, whole	1 lg	189	136
Dried peaches	1 c.	139	46
Sweet potatoes	1 c.	134	43
Russet potato, baked	1 lg	123	93
Whole-grain bread	1 slice	36	46
Broccoli	½ c.	34	24
Cauliflower	½ c.	16	13

Magnesium (mg)

Food	Serv	mg	%DV
Almonds	1 oz	80	20
Cashews	1 oz	74	19
Soy milk	1 c.	61	15
Black beans	½ c.	60	15
Edamame	½ c.	50	13
Bread	2 slices	46	12
Avocado	1 c.	44	11
Brown rice	½ c.	42	11
Yogurt	8 oz	42	11
Oatmeal, instant	1 packet	36	9
Salmon	3 oz	26	7
Chicken breast	3 oz	22	6
Apple	1 med	9	2

TRACE MINERALS

Iron (mg)

Food	Serv	mg	%DV
Cereals, fortified	1 serv	18	100
Oysters	3 oz	8	44
Dark chocolate	3 oz	7	39
Beef liver	3 oz	5	28
Lentils	½ c.	3	17
Spinach, boiled	½ c.	3	17
Tofu, firm	½ c.	3	17
Kidney beans	½ c.	2	11
Sardines	3 oz	2	11

Zinc (mg)

Food	Serv	mg	%DV
Oysters	3 oz	74	493
Beef, chuck roast	3 oz	7	47
Crab	3 oz	6.5	43
Lobster	3 oz	3.4	23
Pork loin	3 oz	2.9	19
Baked beans	½ c.	2.9	19
Yogurt, low fat	8 oz	1.7	11
Oatmeal, instant	1 packet	1.1	7
Almonds	1 oz	0.9	6

Selenium (mcg)

Food	Serv	mcg	%DV
Brazil nuts	1 oz	544	777
Couscous	1 c.	43	61
Crab meat	3 oz	41	59
Ricotta cheese	1 c.	41	59
Salmon	3 oz	40	57
Pork	3 oz	35	50
Shrimp	3 oz	34	49
Round steak	3 oz	28.5	41
Beef liver	3 oz	28	40
Whole-wheat bread	2 slices	23	33
Chicken	3 oz	13	19
Barley, cooked	1 c.	13.5	19
Ground beef	3 oz	18	26
Milk, low-fat	1 c.	8	11
Walnuts, black	1 oz	5	7

Iodine (mcg)

Food	Serv	mcg	%DV
Seaweed	1 g	16–2,984	11–1,989
Cod fish	3 oz	99	66
Yogurt, low fat	8 oz	75	50
Iodized salt	1.5 g	71	47
Milk, reduced fat	8 oz	56	37
Ice cream, chocolate	½ c.	30	20
Egg	1 lg	24	16
Tuna, canned	3 oz	17	11
Prunes, dried	5 prunes	13	9
Banana	1 med	3	2

Fluoride (mg)

Food	Serv	mg	%DV
Fruit juice	3.5 oz	0.02–2.1	0.7–70
Crab, canned	3.5 oz	0.21	7
Rice, cooked	3.5 oz	0.04	1.3
Fish, cooked	3.5 oz	0.02	0.7
Chicken	3.5 oz	0.015	0.5

Source: Titchenal A. et al., Human Nutrition: 2020 Edition, University of Hawai'i at Manoa (CC BY-NC-SA 4.0). Underlying food values from USDA databases & NIH Office of Dietary Supplements. Ranges (a–b) reflect variation across brands/preparations. This is a condensed extract of the food-value tables only.